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## Degree Correlation and Assortative Mixing

Simon Frost 2026-05-13

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## Introduction

In the standard configuration model, the degree of a node's neighbor is independent of its own degree — the network is neutral (uncorrelated). Real networks, however, exhibit degree correlations:

- Assortative mixing: high-degree nodes preferentially connect to other high-degree nodes (common in social networks).
- Disassortative mixing: high-degree nodes preferentially connect to low-degree nodes (common in biological and technological networks).

These correlations profoundly affect epidemic dynamics. Assortative mixing tends to confine epidemics within degree classes, while disassortative mixing facilitates cross-class transmission.

The key quantity is the mixing matrix  $Q(l|k)$ : the probability that a neighbor of a degree- $k$  node has degree  $l$ . The `CorrelatedPGF` type in `EdgeBasedModels.jl` captures this structure and enables computation of  $R_0$  via the spectral radius of the next-generation matrix.

References: Newman (2002) *Phys. Rev. Lett.* 89, 208701; Wang et al. (2019) *J. Math. Biol.* 78, 1923–1951.

## Setup

```
using EdgeBasedModels
using LinearAlgebra
using Plots
```

## Mixing Matrices

The conditional probability  $Q(l|k)$  defines the degree correlation structure. For a network with maximum degree  $K$  and degree distribution  $\{p_k\}$ :

- Neutral mixing:  $Q(l|k) = \frac{lp_l}{\langle k \rangle}$ , independent of  $k$ . A degree- $l$  neighbor is chosen proportional to  $lp_l$  (excess degree distribution).
- Assortative mixing (Newman):  $Q_r(l|k) = r \delta_{k,l} + (1 - r) Q_{\text{neutral}}(l|k)$  where  $r \in [0, 1]$ .
  - $r = 0$ : neutral (uncorrelated)
  - $r = 1$ : perfectly assortative (every neighbor has the same degree as the ego node)

The next-generation matrix for disease spread on a degree-correlated network is:

$$C_{k,l} = (k-1)Q(l|k)$$

and  $R_0 = T \cdot \rho(C)$ , where  $T$  is the transmissibility and  $\rho$  denotes the spectral radius.

## Building Correlated Networks

We use a bimodal degree distribution — 50% degree-3 and 50% degree-7 — to make the degree correlation structure clearly visible.

```
# Degree distribution: p_k for k = 0, 1, ..., 7
probs = zeros(8)
probs[4] = 0.5 # k = 3 (index = k+1)
probs[8] = 0.5 # k = 7

# Verify
mean_k = sum((k - 1) * probs[k] for k in eachindex(probs))
var_k = sum((k - 1)^2 * probs[k] for k in eachindex(probs)) - mean_k^2
println("Mean degree: ", mean_k)
println("Variance:    ", var_k)
```

```
Mean degree: 5.0
Variance:    4.0
```

Now construct neutral, mildly assortative, and strongly assortative networks:

```
neutral = neutral_correlated_pgf(probs)
mild     = assortative_correlated_pgf(probs, 0.3)
strong   = assortative_correlated_pgf(probs, 0.8)
```

```
CorrelatedPGF(DegreePGF(z, 0.5(z^3 + z^7)), 7, [0.0, 0.0, 0.0, 0.5, 0.0, 0.0, 0.0, 0.5], [0.8 0
```

## Visualising mixing matrices

Each row of the mixing matrix  $Q$  gives the neighbor-degree distribution for nodes of a given degree. For the bimodal distribution, only rows and columns for  $k = 3$  and  $k = 7$  carry weight.

```

degrees = 0:7
p1 = heatmap(degrees, degrees, neutral.mixing_matrix',
             xlabel = "Ego degree k", ylabel = "Neighbor degree l",
             title = "Neutral (r=0)", clims = (0, 1), color = :viridis, aspect_ratio = 1)
p2 = heatmap(degrees, degrees, mild.mixing_matrix',
             xlabel = "Ego degree k", ylabel = "Neighbor degree l",
             title = "Mild (r=0.3)", clims = (0, 1), color = :viridis, aspect_ratio = 1)
p3 = heatmap(degrees, degrees, strong.mixing_matrix',
             xlabel = "Ego degree k", ylabel = "Neighbor degree l",
             title = "Strong (r=0.8)", clims = (0, 1), color = :viridis, aspect_ratio = 1)
plot(p1, p2, p3, layout = (1, 3), size = (900, 300))

```

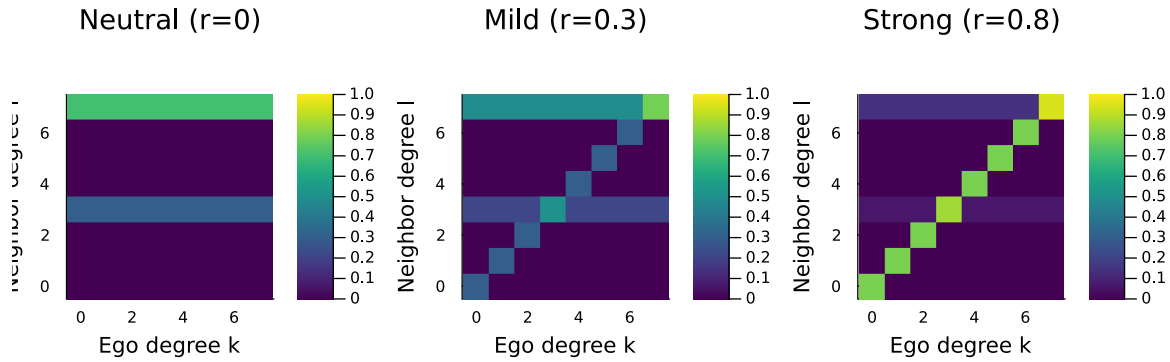


Figure 1: Figure 1: Mixing matrices  $Q(l|k)$  for neutral ( $r=0$ ), mild ( $r=0.3$ ), and strong ( $r=0.8$ ) assortativity. Only degrees 3 and 7 are present.

In the neutral case, the neighbor-degree distribution is the same regardless of ego degree. As  $r$  increases, mass concentrates on the diagonal — degree-3 nodes increasingly connect to degree-3 nodes and degree-7 nodes to degree-7 nodes.

## R and Assortativity

How does assortativity affect  $R_0$ ? We sweep  $r$  from 0 to 1 for a fixed transmissibility  $T = 0.75$  ( $\beta = 0.75, \gamma = 0.25$ ).

```

T = 0.75
r_vals = 0.0:0.02:1.0
R0_bimodal = [correlated_R0 assortative_correlated_pgf(probs, r), T] for r in r_vals]

```

51-element Vector{Float64}:

```

3.5999999999999999
3.610610198112615
3.621444106815887
3.6325067452299065
3.6438031925820478
3.655338583677548
3.667118103903502
3.6791469837509574
3.691430492841646
3.703973933448026
?
4.296241915818795
4.320362858982983
4.344870712990218
4.369765082152217
4.395045270942502
4.420710286693302
4.446758843425922
4.473189366780302
4.5

```

```

plot(r_vals, R0_bimodal, linewidth = 2, color = :darkred, label = "Bimodal (k=3,7)",
     xlabel = "Assortativity parameter r", ylabel = "R0",
     title = "R0 vs Assortativity (T = $T)")
hline!([1.0], linestyle = :dash, color = :gray, label = "R0 = 1")

```

For the bimodal distribution, increasing assortativity decreases  $R_0$ . Intuitively, assortative mixing isolates the high-degree nodes into a cluster that saturates quickly, reducing their ability to seed the epidemic into the broader network.

## Comparing degree distributions

The effect of assortativity depends on the shape of the degree distribution. We compare three distributions with similar mean degrees:

```

# Poisson-like (truncated at k=15) with mean ≈ 5
poisson_probs = zeros(16)
for k in 0:15
    poisson_probs[k + 1] = exp(-5.0) * 5.0^k / factorial(k)
end
poisson_probs ./= sum(poisson_probs)

```

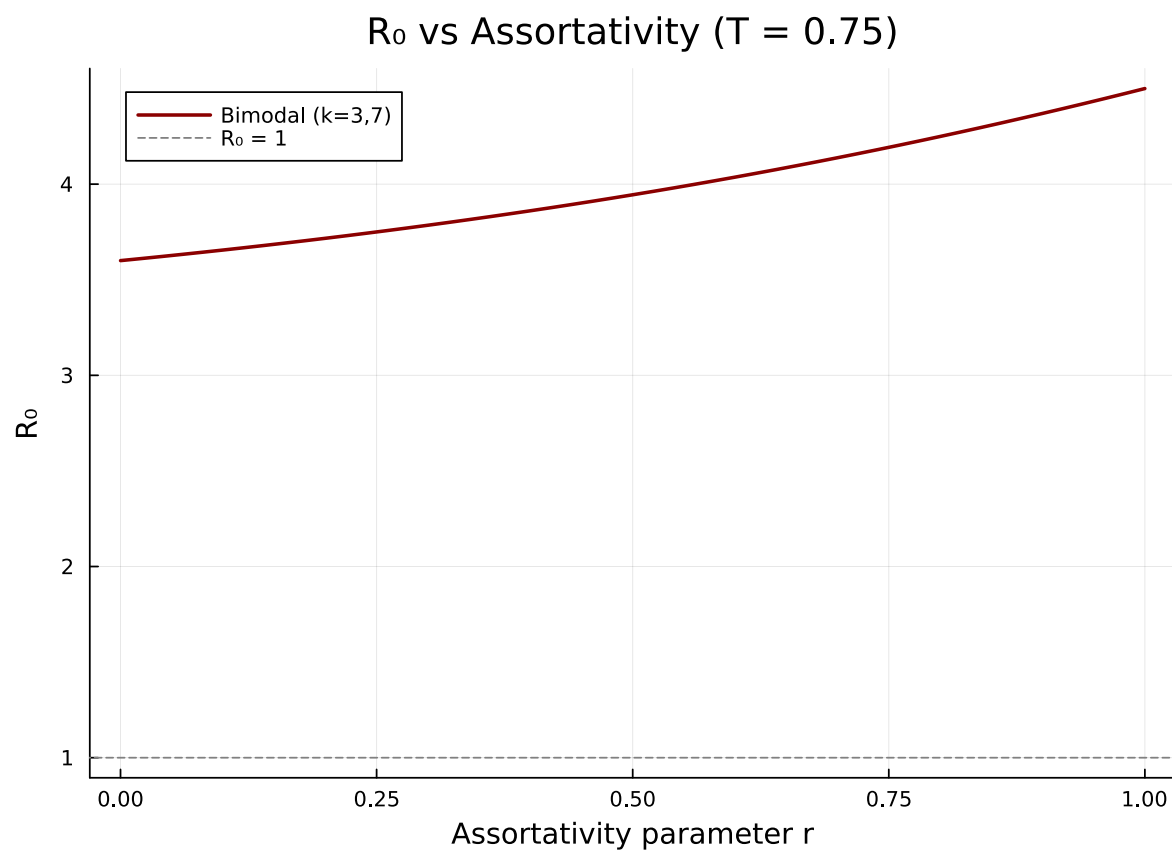


Figure 2: Figure 2:  $R_0$  as a function of assortativity parameter  $r$  for a bimodal degree distribution. Assortativity reduces  $R_0$ .

```

# Power-law-like (truncated):  $p_k \propto k^{-2}$  for  $k = 1..15$ 
pl_probs = zeros(16)
for k in 1:15
    pl_probs[k + 1] = k^(-2.0)
end
pl_probs ./= sum(pl_probs)

println("Poisson-like mean: ", round(sum((k - 1) * poisson_probs[k] for k in eachindex(poisson_probs)), 1))
println("Power-law-like mean: ", round(sum((k - 1) * pl_probs[k] for k in eachindex(pl_probs)), 1))

```

Poisson-like mean: 5.0  
Power-law-like mean: 2.1

```

R0_poisson = [correlated_R0(assortative_correlated_pgf(poisson_probs, r), T) for r in r_vals]
R0_pl = [correlated_R0(assortative_correlated_pgf(pl_probs, r), T) for r in r_vals]

```

```

51-element Vector{Float64}:
 2.640362757650594
 2.708689352348886
 2.7797268170247955
 2.8536217704533557
 2.930528246790755
 3.0106076766308525
 3.09402873842283
 3.180967041511214
 3.271604592810858
 3.3661289879000647
 3.461804592810858
 3.55860027066869815
 3.6564129459101377
 3.755268550191887073
 3.85473252006413386
 3.955678202940233419
 4.0583375470169886
 4.16288745805968028
 4.2694293307086896
 4.3778105

```

```

plot(r_vals, R0_bimodal, linewidth = 2, label = "Bimodal (k=3,7)", color = :darkred,
     xlabel = "Assortativity parameter r", ylabel = "R0",
     title = "R0 vs Assortativity for Different Distributions (T = $T)")
plot!(r_vals, R0_poisson, linewidth = 2, label = "Poisson-like (mean≈5)", color = :blue)
plot!(r_vals, R0_pl, linewidth = 2, label = "Power-law-like", color = :purple)
hline!([1.0], linestyle = :dash, color = :gray, label = "R0 = 1")

```

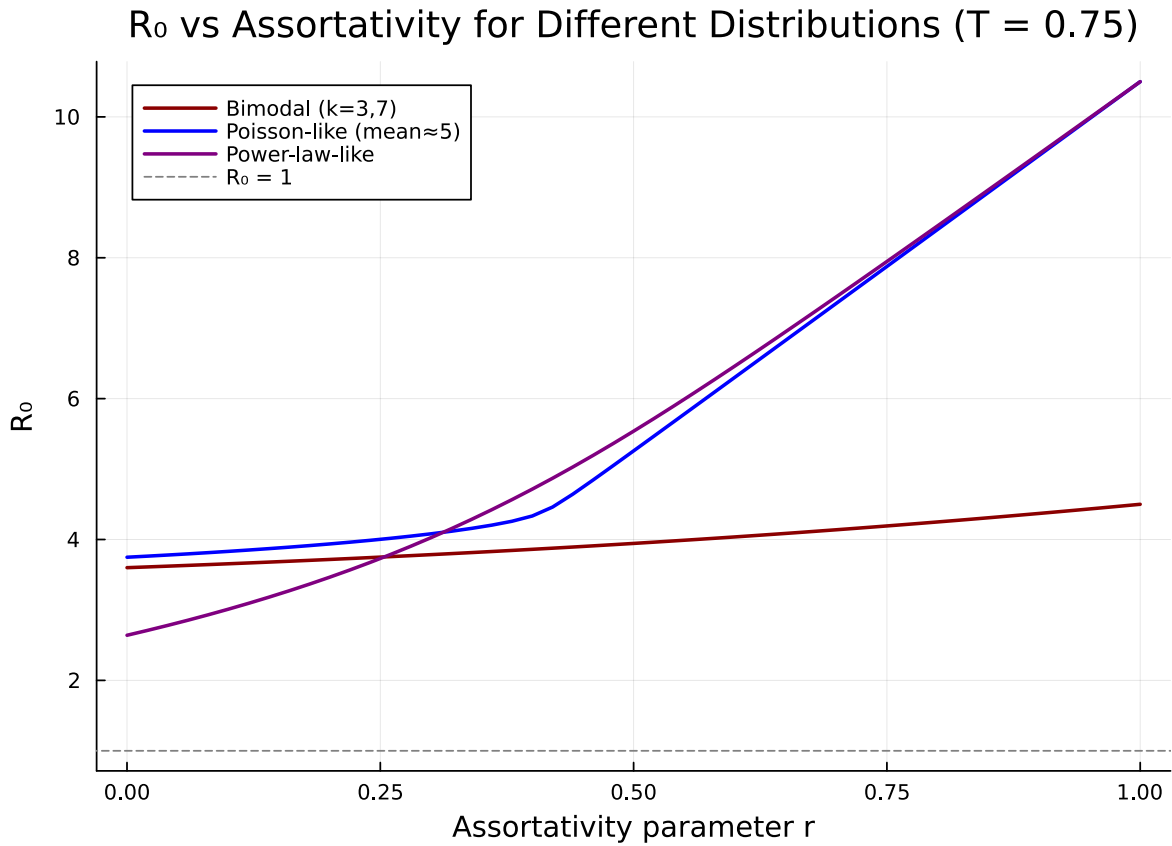


Figure 3: Figure 3:  $R_0$  vs assortativity for different degree distributions. The effect depends on the shape of the distribution.

## Spectral Analysis

$R_0$  is determined by the spectral radius of the next-generation matrix  $C_{k,l} = (k-1)Q(l|k)$ . Examining the eigenvalue spectrum reveals how assortativity reshapes transmission pathways.



```

function next_gen_matrix(cpgf::CorrelatedPGF)
    K = cpgf.max_degree
    C = zeros(K + 1, K + 1)
    for k in 0:K, l in 0:K
        C[k + 1, l + 1] = max(k - 1, 0) * cpgf.mixing_matrix[k + 1, l + 1]
    end
    return C
end

C_neutral = next_gen_matrix(neutral)
C_mild    = next_gen_matrix(mild)
C_strong  = next_gen_matrix(strong)

eig_neutral = sort(abs.(eigvals(C_neutral))); rev = true)
eig_mild    = sort(abs.(eigvals(C_mild))); rev = true)
eig_strong  = sort(abs.(eigvals(C_strong))); rev = true)

```

```

8-element Vector{Float64}:
 5.665547783358538
 4.0
 3.2
 2.4000000000000004
 1.6944522166414628
 0.8
 0.0
 0.0

```

```

bar_width = 0.25
n = length(eig_neutral)
xs = 1:n
bar(xs .- bar_width, eig_neutral, bar_width = bar_width, label = "Neutral (r=0)", color =
bar!(xs, eig_mild, bar_width = bar_width, label = "Mild (r=0.3)", color = :orange, alpha =
bar!(xs .+ bar_width, eig_strong, bar_width = bar_width, label = "Strong (r=0.8)", color =
xlabel!("Eigenvalue index")
ylabel!("|λ|")
title!("Eigenvalue Spectrum of Next-Generation Matrix")

```

The leading eigenvalue (spectral radius) decreases with assortativity. For the bimodal distribution, assortative mixing splits the single dominant eigenvalue into two smaller ones corresponding to within-class transmission for each degree class.

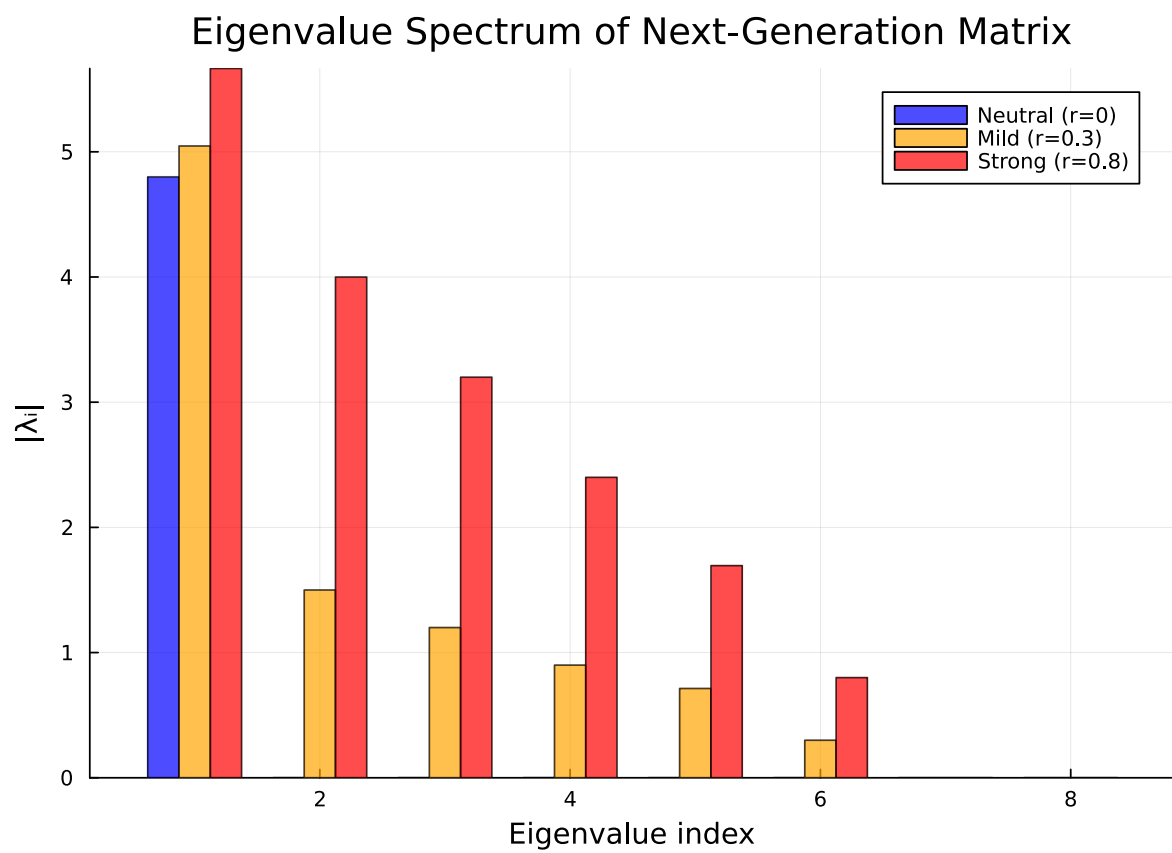


Figure 4: Figure 4: Eigenvalue spectrum of the next-generation matrix for different levels of assortativity.

```
println("Spectral radius (neutral): ", round(eig_neutral[1]; digits = 4))
println("Spectral radius (mild):    ", round(eig_mild[1]; digits = 4))
println("Spectral radius (strong): ", round(eig_strong[1]; digits = 4))
```

```
Spectral radius (neutral): 4.8
Spectral radius (mild):    5.0467
Spectral radius (strong):  5.6655
```

## Comparison with Uncorrelated Model

For neutral mixing, the correlated model should recover the standard (uncorrelated)  $R_0$ :

$$R_0 = T \cdot \frac{\langle k^2 \rangle - \langle k \rangle}{\langle k \rangle}$$

Let us verify this numerically.

```
# Standard uncorrelated R0 from moments
function uncorrelated_R0(degree_probs, T)
    mean_k = sum((k - 1) * degree_probs[k] for k in eachindex(degree_probs))
    mean_k2 = sum((k - 1)^2 * degree_probs[k] for k in eachindex(degree_probs))
    return T * (mean_k2 - mean_k) / mean_k
end

for (name, dp) in [("Bimodal", probs), ("Poisson-like", poisson_probs), ("Power-law-like",
    r0_corr = correlated_R0(neutral_correlated_pgf(dp), T)
    r0_uncorr = uncorrelated_R0(dp, T)
    println("$name:")
    println("  Correlated (neutral): ", round(r0_corr; digits = 6))
    println("  Uncorrelated formula: ", round(r0_uncorr; digits = 6))
    println("  Difference:           ", round(abs(r0_corr - r0_uncorr); digits = 10))
    println()
end
```

Bimodal:

```
Correlated (neutral): 3.6
Uncorrelated formula: 3.6
Difference:           0.0
```

Poisson-like:

```

Correlated (neutral): 3.748231
Uncorrelated formula: 3.748231
Difference:           0.0

```

Power-law-like:

```

Correlated (neutral): 2.640363
Uncorrelated formula: 2.640363
Difference:           0.0

```

The agreement is exact (up to floating-point precision), confirming that neutral CorrelatedPGF is consistent with the standard uncorrelated theory.

### Divergence with assortativity

As  $r$  increases, the correlated  $R_0$  diverges from the uncorrelated value:

```

r_sweep = 0.0:0.05:1.0
r0_uncorr_bimodal = uncorrelated_R0(probs, T)

diff_bimodal = [correlated_R0(assortative_correlated_pgf(probs, r), T) - r0_uncorr_bimodal
                for r in r_sweep]
diff_poisson = [correlated_R0(assortative_correlated_pgf(poisson_probs, r), T) - uncorrelated_R0(poisson_probs, T)
                for r in r_sweep]

```

21-element Vector{Float64}:

```

4.440892098500626e-16
0.03931160621469676
0.0831088666368176
0.13242433129557485
0.1887175206846914
0.25419300928923505
0.33257105263211884
0.4317339708090353
0.5856212229478546
0.9949742024831396
␣
2.555491891405655
3.0795566713283176
3.603895283533646
4.128386459108504
4.652971283055713

```

5.177617701529076  
5.7023067855073  
6.227026639075639  
6.751769411471962

```
plot(r_sweep, diff_bimodal, linewidth = 2, label = "Bimodal", color = :darkred,
     marker = :circle, markersize = 3,
     xlabel = "Assortativity parameter r",
     ylabel = " $R_0(\text{correlated}) - R_0(\text{uncorrelated})$ ",
     title = "Deviation from Uncorrelated  $R_0$ ")
plot!(r_sweep, diff_poisson, linewidth = 2, label = "Poisson-like", color = :blue,
      marker = :square, markersize = 3)
hline!([0.0], linestyle = :dash, color = :gray, label = nothing)
```

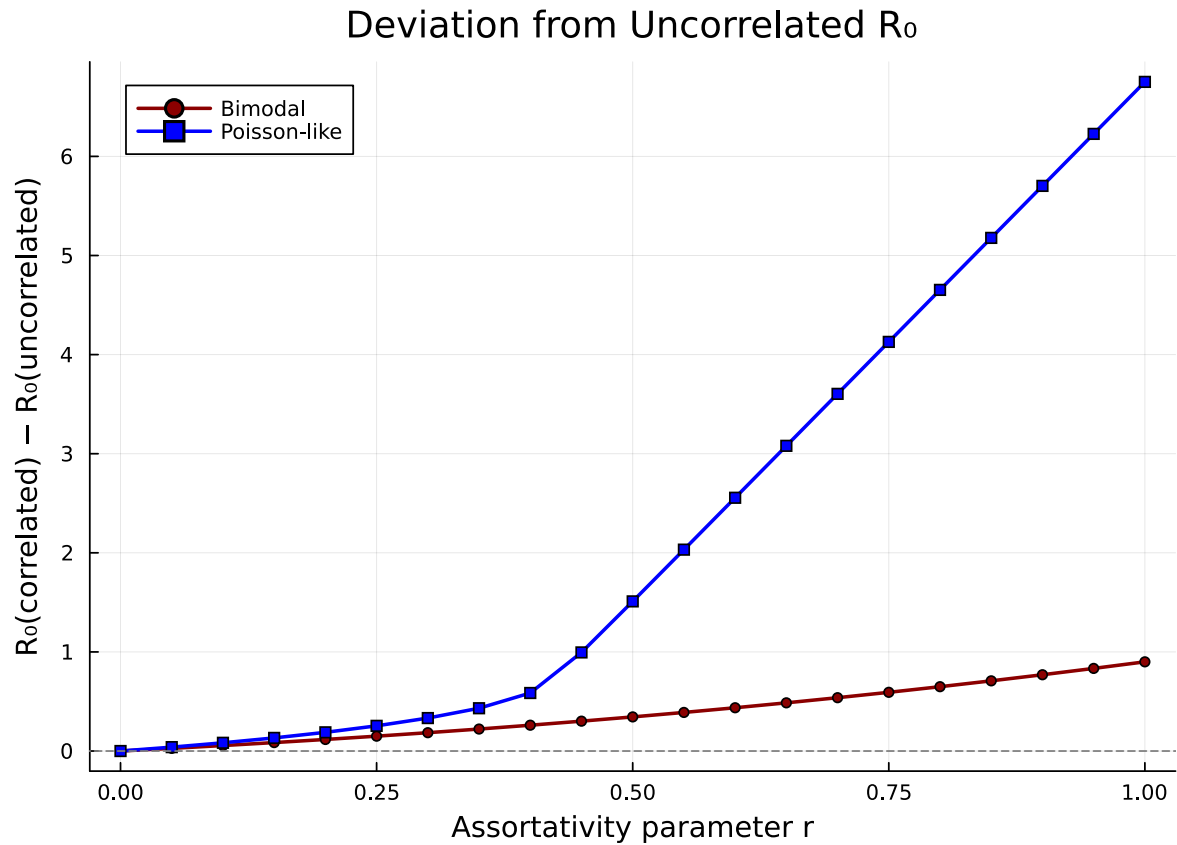


Figure 5: Figure 5: Difference between correlated  $R_0$  and the standard uncorrelated  $R_0$  as assortativity increases.

The uncorrelated formula overestimates  $R_0$  for assortative networks — degree correlations matter,

and ignoring them leads to overly pessimistic epidemic predictions.

## Custom Mixing Matrices

Not all mixing patterns follow the simple assortative interpolation. Disassortative networks, where high-degree nodes preferentially connect to low-degree nodes, are common in sexual contact networks and bipartite-like structures.

We construct a disassortative mixing matrix by flipping the assortative pattern — concentrating off-diagonal mass:

```
function disassortative_mixing(degree_probs, r_disassort)
    K = length(degree_probs) - 1
    neutral_cpgf = neutral_correlated_pgf(degree_probs)
    Q_neutral = neutral_cpgf.mixing_matrix

    # Anti-diagonal concentration: prefer neighbors of maximally different degree
    Q = copy(Q_neutral)
    for k in 1:K+1, l in 1:K+1
        # Weight toward opposite end of degree spectrum
        anti_diag = abs(k - l) / K
        Q[k, l] = (1 - r_disassort) * Q_neutral[k, l] + r_disassort * anti_diag * Q_neutral[k, l]
    end
    # Re-normalise rows
    for k in 1:K+1
        s = sum(Q[k, :])
        if s > 1e-12
            Q[k, :] ./= s
        end
    end
    return correlated_pgf(degree_probs, Q)
end
```

disassortative\_mixing (generic function with 1 method)

```
disassort = disassortative_mixing(probs, 0.8)
```

CorrelatedPGF(DegreePGF(z, 0.5(z<sup>3</sup> + z<sup>7</sup>)), 7, [0.0, 0.0, 0.0, 0.5, 0.0, 0.0, 0.0, 0.5], [0.0 0

```

degrees = 0:7
p1 = heatmap(degrees, degrees, strong.mixing_matrix',
             xlabel = "Ego degree k", ylabel = "Neighbor degree l",
             title = "Assortative (r=0.8)", clims = (0, 1), color = :viridis, aspect_ratio = 1)
p2 = heatmap(degrees, degrees, disassort.mixing_matrix',
             xlabel = "Ego degree k", ylabel = "Neighbor degree l",
             title = "Disassortative", clims = (0, 1), color = :viridis, aspect_ratio = 1)
plot(p1, p2, layout = (1, 2), size = (700, 300))

```

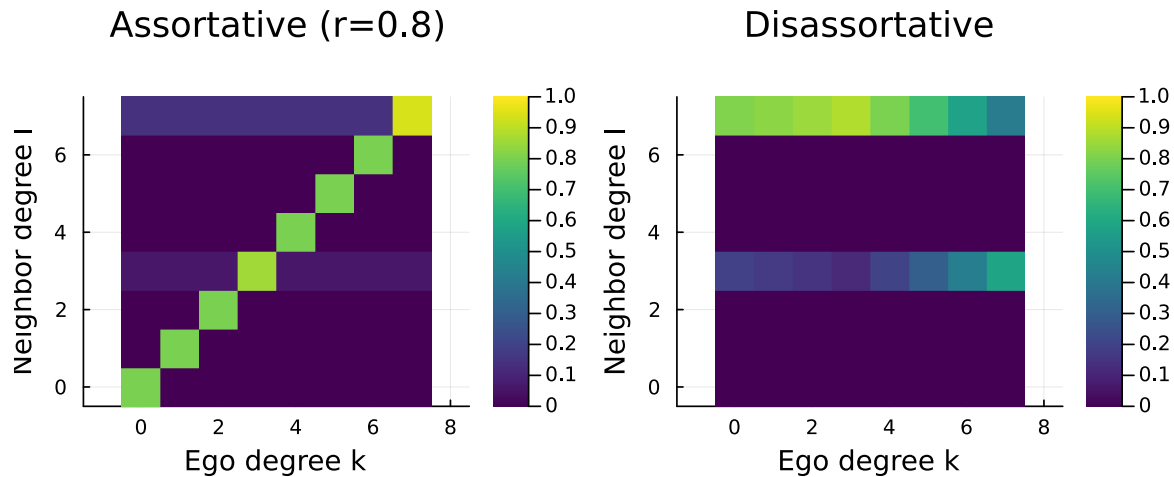


Figure 6: Figure 6: Mixing matrix for a disassortative network — high-degree nodes preferentially connect to low-degree nodes.

```

println("R0 (neutral):      ", round(correlated_R0(neutral, T); digits = 4))
println("R0 (assortative):    ", round(correlated_R0(strong, T); digits = 4))
println("R0 (disassortative):", round(correlated_R0(disassort, T); digits = 4))

```

```

R0 (neutral):      3.6
R0 (assortative):  4.2492
R0 (disassortative):3.0728

```

Disassortative mixing can increase  $R_0$  relative to neutral mixing because high-degree nodes efficiently bridge to many low-degree nodes, creating broad transmission cascades. This is relevant for modelling sexually transmitted infections, where highly active individuals connect to less active ones.

## Summary

- Degree correlations (who connects to whom based on degree) fundamentally alter epidemic dynamics beyond what the degree distribution alone predicts.
- The mixing matrix  $Q(l|k)$  captures the conditional neighbor-degree distribution and is the core input to `CorrelatedPGF`.
- Assortative mixing ( $r > 0$ ) tends to reduce  $R_0$  by confining epidemics within degree classes.
- Disassortative mixing can increase  $R_0$  by enabling efficient cross-class transmission.
- For neutral mixing ( $r = 0$ ), the correlated  $R_0$  exactly recovers the standard formula  $R_0 = T(\langle k^2 \rangle - \langle k \rangle) / \langle k \rangle$ .
- `EdgeBasedModels.jl` provides `neutral_correlated_pgf`, `assortative_correlated_pgf`, `correlated_pgf` (custom matrix), and `correlated_R0` for analysing degree-correlated networks.

## Simulation validation

This vignette focuses on a scenario for which `NetworkOutbreaks.jl` does not yet provide an out-of-the-box stochastic ground truth (multi-type host graphs with prescribed mixing matrices, time-varying networks via the EBCM rewiring schedule, multiplex layers, degree-correlated configuration models, clustered graphs with prescribed triangle counts, or final-size sweeps over  $R_0$ ).

A future revision will add the missing primitives to `NetworkOutbreaks.jl` and overlay the corresponding Gillespie SSA ribbon here. Until then, see [vignette 01](#) for the validation pattern on a single-layer Poisson configuration-model network with the same canonical parameters.